

Integration by Parts

ANSWER KEY



The integration by parts formula

- 1 Evaluate $\int x e^x dx$ using integration by parts.
Let $u = x$, $dv = e^x dx$: $= x e^x - \int e^x dx = x e^x - e^x + C$.
- 2 Evaluate $\int x \cos x dx$ using integration by parts.
Let $u = x$, $dv = \cos x dx$: $= x \sin x - \int \sin x dx = x \sin x + \cos x + C$.
- 3 Evaluate $\int \ln x dx$ using integration by parts (with $dv = dx$).
Let $u = \ln x$, $dv = dx$: $= x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C$.

Repeated and definite applications

- 4 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
First pass ($u = x^2$): $x^2 e^x - 2 \int x e^x dx$. Second pass gives $\int x e^x dx = x e^x - e^x$. Total:
 $x^2 e^x - 2 x e^x + 2 e^x + C$.
- 5 Evaluate $\int_0^1 x e^{2x} dx$ exactly.
Let $u = x$, $dv = e^{2x} dx$: $\left[\frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} \right]_0^1 = \left(\frac{e^2}{2} - \frac{e^2}{4} \right) - \left(0 - \frac{1}{4} \right) = \frac{e^2}{4} + \frac{1}{4}$.

Choosing u and dv wisely (LIATE)

- 6 Evaluate $\int x \ln x dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).
 $u = \ln x$, $dv = x dx$, so $du = \frac{1}{x} dx$, $v = \frac{x^2}{2}$: $= \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$.
- 7 Evaluate $\int x^2 \cos x dx$, applying integration by parts twice.
First: $u = x^2$, $dv = \cos x dx$: $x^2 \sin x - 2 \int x \sin x dx$. Second (on $\int x \sin x dx$): $= -x \cos x + \sin x$. Total:
 $x^2 \sin x + 2 x \cos x - 2 \sin x + C$.

The recurring-integral trick

- 8 Evaluate $\int e^x \sin x dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.
Let $I = \int e^x \sin x dx$. First pass ($u = \sin x$, $dv = e^x dx$): $I = e^x \sin x - \int e^x \cos x dx$. Second pass on the new integral ($u = \cos x$): $\int e^x \cos x dx = e^x \cos x + \int e^x \sin x dx = e^x \cos x + I$. So
 $I = e^x \sin x - e^x \cos x - I$, giving $2I = e^x (\sin x - \cos x)$, so $I = \frac{e^x (\sin x - \cos x)}{2} + C$.

Definite integrals by parts

- 9 Evaluate $\int_0^{\pi} x \sin x \, dx$ exactly.

$$u = x, dv = \sin x \, dx: [-x \cos x + \sin x]_0^{\pi} = (-\pi \cos \pi + \sin \pi) - (0) = \pi.$$

- 10 Evaluate $\int_1^e x^2 \ln x \, dx$ exactly.

$$u = \ln x, dv = x^2 \, dx: \left[\frac{x^3}{3} \ln x - \frac{x^3}{9} \right]_1^e = \left(\frac{e^3}{3} - \frac{e^3}{9} \right) - \left(0 - \frac{1}{9} \right) = \frac{2e^3}{9} + \frac{1}{9} = \frac{2e^3 + 1}{9}.$$

Reduction formulas

- 11 Use integration by parts with $u = x^n$, $dv = e^x \, dx$ to derive the reduction formula

$$\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx.$$

$$du = nx^{n-1} \, dx, v = e^x: \int x^n e^x \, dx = x^n e^x - \int e^x \cdot nx^{n-1} \, dx = x^n e^x - n \int x^{n-1} e^x \, dx.$$

- 12 Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x \, dx$, confirming it matches an earlier result.

$$\int x^2 e^x \, dx = x^2 e^x - 2 \int x e^x \, dx. \text{ Applying again: } \int x e^x \, dx = x e^x - \int e^x \, dx = x e^x - e^x. \text{ Total:}$$

$$x^2 e^x - 2(x e^x - e^x) = x^2 e^x - 2x e^x + 2e^x + C, \text{ matching the earlier double-integration-by-parts result.}$$

Integration by parts with inverse trig functions

- 13 Evaluate $\int \tan^{-1} x \, dx$ using integration by parts with $u = \tan^{-1} x$, $dv = dx$.

$$du = \frac{1}{1+x^2} \, dx, v = x: = x \tan^{-1} x - \int \frac{x}{1+x^2} \, dx = x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C.$$

- 14 Evaluate $\int x \sin^{-1} x \, dx$ using $u = \sin^{-1} x$, $dv = x \, dx$ (the resulting integral can be left in terms of a known antiderivative form).

$$du = \frac{1}{\sqrt{1-x^2}} \, dx, v = \frac{x^2}{2}: = \frac{x^2}{2} \sin^{-1} x - \frac{1}{2} \int \frac{x^2}{\sqrt{1-x^2}} \, dx, \text{ which requires a further trigonometric substitution to complete.}$$

A capstone mixed-technique problem

- 15 Evaluate $\int_0^{\pi/2} x \cos(2x) \, dx$ exactly, using integration by parts.

$$u = x, dv = \cos(2x) \, dx, v = \frac{1}{2} \sin(2x):$$

$$\left[\frac{x}{2} \sin(2x) \right]_0^{\pi/2} - \frac{1}{2} \int_0^{\pi/2} \sin(2x) \, dx = \left(\frac{\pi}{4} \sin \pi - 0 \right) - \frac{1}{2} \left[-\frac{1}{2} \cos(2x) \right]_0^{\pi/2} = 0 - \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = -\frac{1}{2}.$$

Visualizing an area found by integration by parts

- 16 The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.

$$u = x, dv = e^{-x} dx, v = -e^{-x}:$$

$$[-xe^{-x}]_0^3 + \int_0^3 e^{-x} dx = -3e^{-3} + [-e^{-x}]_0^3 = -3e^{-3} + (1 - e^{-3}) = 1 - 4e^{-3} \approx 0.801.$$

Choosing between substitution and parts

- 17 For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u -substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2} dx$; (b) $\int xe^{2x} dx$.
- (a) Substitution — $u = x^2$ makes $du = 2x dx$ appear directly, no product rule needed. (b) Integration by parts — x and e^{2x} are genuinely different function types with no simplifying substitution; use $u = x$, $dv = e^{2x} dx$.
- 18 Evaluate $\int \sec^3 x dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.
- $du = \sec x \tan x dx$, $v = \tan x$:
- $$I = \sec x \tan x - \int \sec x \tan^2 x dx = \sec x \tan x - \int \sec x (\sec^2 x - 1) dx = \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$
- . So $2I = \sec x \tan x + \ln|\sec x + \tan x|$, giving $I = \frac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|) + C$.
- 19 A city's population growth rate is modeled by $r(t) = t \ln(t + 1)$ thousand people per year, for $0 \leq t \leq 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.
- Total growth $= \int_0^5 t \ln(t + 1) dt$. This does not integrate cleanly with $u = \ln(t + 1)$, $dv = t dt$ alone; using $u = \ln(t + 1)$, $dv = t dt$, $v = \frac{t^2}{2}$: $\left[\frac{t^2}{2} \ln(t + 1)\right]_0^5 - \frac{1}{2} \int_0^5 \frac{t^2}{t + 1} dt$. Dividing $\frac{t^2}{t + 1} = t - 1 + \frac{1}{t + 1}$:
- $$\frac{1}{2} \int_0^5 \left(t - 1 + \frac{1}{t + 1}\right) dt = \frac{1}{2} \left[\frac{t^2}{2} - t + \ln(t + 1)\right]_0^5 = \frac{1}{2}(12.5 - 5 + \ln 6) \approx 4.65.$$
- So total growth $\approx \frac{25}{2} \ln 6 - 4.65 \approx 22.41 - 4.65 \approx 17.76$ thousand people.